

Collated Components:

Bill Of Materials							
Item	Reference(s)	Part #	Value	Description	Qty	Price	Vendor Link
1	C1	GRM31CD80J107MEA8L	100uF	Polarized capacitor, US symbol	1	\$ 0.42	https://www.digikey.com/en/products/detail/murata-electronics/GRM31CD80J107MEA8L/13904781
2	C2	GRM32ER60G337ME05L	330uF	Polarized capacitor, US symbol	1	\$ 1.67	https://www.digikey.com/en/products/detail/murata-electronics/GRM32ER60G337ME05L/6155746
3	C3, C4, C5	GRM033C71C104KE14D	0.1uF	capacitor, US symbol	3	\$ 0.24	https://www.digikey.com/en/products/detail/murata-electronics/GRM033C71C104KE14D/5027472
4	D1, D3	L1MC-RGB0028000MP0	Green LED, Red LED	Light emitting diode	2	\$ 1.02	https://www.digikey.com/en/products/detail/lumileds/L1MC-RGB0028000MP0/18071971
5	D2	1N5819HW-7-F	1N5819	40V 1A Schottky Barrier Rectifier Diode, DO-41	1	\$ 0.29	https://www.digikey.com/en/products/detail/diodes-incorporated/1N5819HW-7-F/814970
6	F1		Fuse	Fuse	1		PRLTA 10
7	J1	USB3131-30-0230-A	USB_B_Micro	USB Micro Type B connector	1	\$ 0.78	https://www.digikey.com/en/products/detail/gct/USB3131-30-0230-A/9859642
8	J2, J4		Conn_02x04	Generic connector, double row, 02x04, counter clockwise	1		PRLTA 10
9	J3	PJ-102AH	Barrel_Jack_Switch	DC Barrel Jack with an internal switch	1	\$ 0.76	https://www.amazon.com/ELEGOO-Electronics-Potentiometer-tie-Points-Breadboard/dp/B09YRJQRFF
10	J5		Extra Headers	Generic connector, single row, 01x04, script generated	1		PRLTA 10
11	JP1, JP2, JP3		Jumper_2_Bridged	Jumper, 2-pole, closed/bridged	3		PRLTA 10
12	L1		330uH	Inductor	1		PRLTA 10
13	R1, R2, R3, R5, R6	RC0402JR-0710KL	10k	Resistor, US symbol	5	\$ 0.50	https://www.digikey.com/en/products/detail/yageo/RC0402JR-0710KL/726418
14	R4	PFR05S-221-FNH	220	Resistor, US symbol	1	\$ 0.10	https://www.digikey.com/en/products/detail/delta-electronics-cyntec/PFR05S-221-FNH/9761550
15	SW1	CSMS15CIC06	SW_Push	Push button switch, generic, two pins	2	\$ 14.34	
16	TP1, TP2, TP3, TP4, TP5, TP6, TP7		TestPoint	test point	7		PRLTA 10
17	U1	ESP32-S3-WROOM-1-N4	ESP32-S3-WROOM-1-N4	microcontroller	1	\$ 5.06	https://www.digikey.com/en/products/detail/espressif-systems/ESP32-S3-WROOM-1-N4/16162639
18	U2	LM2575D2T-3.3R4	LM2575-3.3BU	Fixed 3.3V 52kHz Simple 1A Buck Regulator, TO-263	1	\$ 1.08	https://www.digikey.com/en/products/detail/onsemi/LM2575D2T-3-3R4/1476687
Total Price						\$ 26.26	